

INVENTION DISCLOSURE FORM FOR PATENTS

Applicant Name-Marwadi University

1. Particulars of Inventors

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2. Provide title of the invention: AI-Powered Waste Segregation using Computer Vision & Machine Learning

3. In 100 words or less, please provide an abstract or summary of the invention:

This invention proposes an AI-based system for automatic waste segregation using image processing and machine learning techniques. The system analyzes images of waste items and classifies them into predefined categories such as plastic, glass, metal, paper, cardboard, and biodegradable waste. By automating the identification process, the system reduces manual effort, improves segregation accuracy, and enhances recycling efficiency. The proposed solution is scalable, cost-effective, and suitable for real-time deployment in smart waste management systems.

4. Detail description of the invention:(Answer to all below are required in detail)

a. Problem the invention is solving

Manual waste segregation is inefficient, time-consuming, and unsafe, leading to poor recycling outcomes. Improper segregation causes recyclable materials to be contaminated, increasing landfill usage and environmental pollution. Existing systems lack accuracy and real-time performance under diverse conditions.

b. General Utility/application of the invention

- Smart waste management systems
- Recycling plants and material recovery facilities
- Smart city infrastructure
- Educational and awareness systems for waste segregation

c. Advantages of the invention disclosing about the increased efficiency/efficacy

- Reduces human intervention and health risks

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- Improves segregation accuracy and efficiency
- Works on image-based inputs without specialized hardware
- Scalable and adaptable to new waste categories

d. Best way of using the invention as well as possible variants

The system works by capturing or uploading waste images, preprocessing them, and classifying the waste type using a trained model. Variants include integration with live cameras, mobile applications, or smart bins for real-time segregation.

e. Working of invention along with Drawing, schematics and flow diagrams if required with complete explanations

- Image input is provided by the user or camera
- Image preprocessing is performed (resizing, normalization)
- Visual features are extracted automatically
- A trained machine learning model classifies the waste
- Output displays waste category and confidence score

5. Have you conducted Primary Patent Search? Yes / No (if yes, attach the patent search results)

No

6. Existing state-of-the-art and prior arts: (Brief background of the existing knowledge/product/process in the market)

Existing waste segregation systems use manual sorting, sensor-based detection, or basic image classification methods. Many approaches lack robustness, real-time capability, or scalability.

7. List out the known ways about how others have tried to solve the same or similar problems? Indicate the disadvantages of these approaches. In addition, please identify any prior art documentation or other material that explains or provides examples of such prior art efforts.

S. No.	Existing state of art	Drawbacks in existing state of art	Overcome (how your invention is overcoming the drawback)
1	Manual sorting	Unsafe and inaccurate	Fully automated classification
2	Sensor-based systems	High cost, limited categories	Image-based and cost-effective

8. List the Technical features and Elements of the invention along with the Description of your invention from start to end.

- Image preprocessing module
- Feature extraction using computer vision
- Machine learning classification model
- Confidence-based prediction output
- User interface for result display

9. List out the features of your invention which are believed to be new and distinguish them over the closest technology.

- Automated image-based waste classification
- Confidence score for prediction reliability
- Scalable design adaptable to real-world deployment

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10. Has the invention been built or tested or implemented? If yes please provide the Efficiency/Efficacy details of the invention

Yes.

The system has been trained and tested on waste image datasets, achieving high classification accuracy and reliable validation performance.

11. Briefly state when and how you first conceived this idea?

The idea was conceived during academic discussions on environmental sustainability and smart city applications, focusing on automating waste segregation using AI.

12. Have you sold, offered for sale, publicly used or published anything related to this invention? If yes, please briefly explain the dates and circumstances. List those individuals to whom you have revealed your invention. Were non-disclosure documents signed prior to disclosure in each case? Please state any deadlines of which you may be aware for filing an application on this invention.

No, The invention has been used only for academic purposes such as project demonstrations and internal evaluations. No commercial sale or public disclosure has been made.

13. Include any reasons that your invention would not have been obvious to someone of average skill in the art.

The integration of computer vision with machine learning for automated waste classification, combined with confidence-based output and real-time applicability, is not obvious to a person of average skill due to its multidisciplinary design and implementation.

14. Additional comments by inventor (if you want to give more details out of scope of this IDF).

The invention has strong potential for commercialization and integration with smart waste bins and IoT-based systems.

15. Drawings/Flowchart/Table

